



GHULAM ISHAQ KHAN INSTITUTE

of Engineering Sciences and Technology

Faculty of Computer Science and Engineering

CE342L • Computational Methods and Tools Lab

Lab Report #1

Limits, Derivatives, Integration, Convergence & Function Plotting

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Semester: 6th Semester

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Objective

The objective of this lab is to apply fundamental calculus concepts — limits, derivatives, and definite integration — using MATLAB's Symbolic Math Toolbox. Additionally, the lab explores the concept of fixed-point iteration to determine whether an iterative function converges or diverges based on the derivative condition $|g'(p)| < 1$.

By the end of this lab, students should be able to:

- Compute limits symbolically in MATLAB using the `limit()` function.
- Differentiate functions using the `diff()` command.
- Evaluate definite integrals using `int()` with bounds.
- Identify convergent and divergent fixed-point iterations using the derivative test.

1 Task 01 — Computing a Limit

► Question

Compute the following limit using MATLAB:

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$$

► Theory

This is one of the most fundamental limits in mathematics. As $x \rightarrow \infty$, the expression $\left(1 + \frac{1}{x}\right)^x$ approaches the mathematical constant $e \approx 2.71828$. This limit is, in fact, the *definition* of Euler's number e . MATLAB's `limit()` function in the Symbolic Math Toolbox evaluates this analytically.

► MATLAB Code

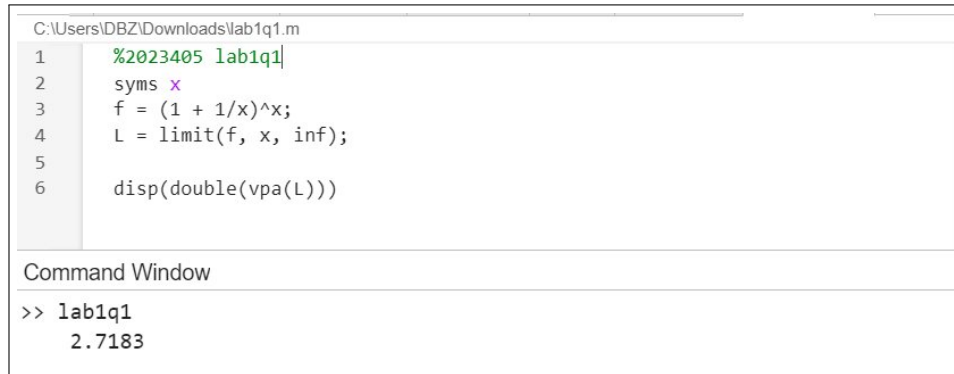
```
1 %2023405 lab1q1
2 syms x
3 f = (1 + 1/x)^x;
4 L = limit(f, x, inf);
```

```

5
6 disp(double(vpa(L)))

```

► Output



The image shows a MATLAB script editor window with the following code:

```

1 %2023405 lab1q1
2 syms x
3 f = (1 + 1/x)^x;
4 L = limit(f, x, inf);
5
6 disp(double(vpa(L)))

```

Below the script editor is the Command Window showing the output:

```

>> lab1q1
    2.7183

```

Figure 1: MATLAB output for Task 01 — Limit computation

► Discussion

The output **2.7183** confirms that the limit evaluates to Euler's number $e \approx 2.71828$. This validates the mathematical identity. MATLAB internally uses symbolic algebra to resolve the indeterminate form, since direct substitution of $x = \infty$ yields the 1^∞ form.

2 Task 02 — Symbolic Differentiation

► Question

Find the derivative of the following function:

$$f(x) = e^{2x} + \sin(x)$$

► Theory

Differentiation is applied term-by-term using standard rules:

$$\frac{d}{dx} [e^{2x}] = 2e^{2x}, \quad \frac{d}{dx} [\sin(x)] = \cos(x)$$

Therefore:

$$f'(x) = 2e^{2x} + \cos(x)$$

► MATLAB Code

```

1 %2023405 lab1q2
2 clc;
3 clear;
4 close all;
5
6 syms x
7 f = exp(2*x) + sin(x);
8 df = diff(f, x)
9
10 pretty(df)

```

► Output



Figure 2: MATLAB output for Task 02 — Derivative of $e^{2x} + \sin(x)$

► Discussion

MATLAB returns $2 \cdot \exp(2 \cdot x) + \cos(x)$, which matches the manually computed result $2e^{2x} + \cos(x)$. The `pretty()` function renders the expression in a more readable mathematical format. This confirms correct application of the chain rule for e^{2x} and the derivative rule for $\sin(x)$.

3 Task 03 — Definite Integration

► Question

Evaluate the following definite integral:

$$\int_0^5 (x^2 + 7x) \, dx$$

► Theory

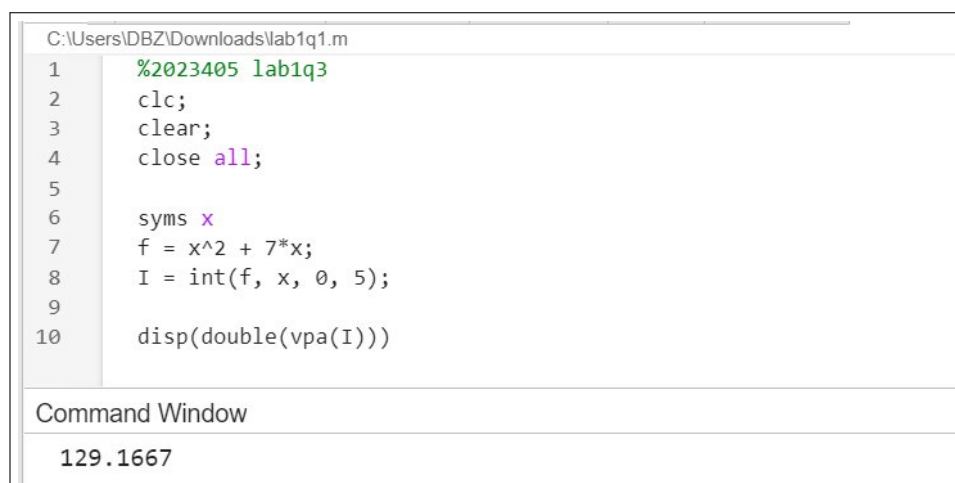
Using the power rule of integration:

$$\int_0^5 (x^2 + 7x) \, dx = \left[\frac{x^3}{3} + \frac{7x^2}{2} \right]_0^5 = \frac{125}{3} + \frac{175}{2} = 41.\overline{66} + 87.5 = 129.\overline{16}$$

► MATLAB Code

```
1 %2023405 lab1q3
2 clc;
3 clear;
4 close all;
5
6 syms x
7 f = x^2 + 7*x;
8 I = int(f, x, 0, 5);
9
10 disp(double(vpa(I)))
```

► Output



```
C:\Users\DBZ\Downloads\lab1q1.m
1 %2023405 lab1q3
2 clc;
3 clear;
4 close all;
5
6 syms x
7 f = x^2 + 7*x;
8 I = int(f, x, 0, 5);
9
10 disp(double(vpa(I)))

Command Window

129.1667
```

Figure 3: MATLAB output for Task 03 — Definite integral result

► Discussion

MATLAB outputs **129.1667**, which corresponds to the exact value $\frac{775}{6} \approx 129.\overline{16}$. This agrees with the manual computation, confirming the correctness of both the symbolic integral and the numerical evaluation via `vpa()`.

4 Task 04 — Fixed-Point Convergence Check

► Question

For the iterative function $g(x) = \frac{x}{2}$, find the fixed point and determine whether the fixed-point iteration is **convergent** or **divergent**.

► Theory

A fixed point p satisfies $g(p) = p$. The iteration $x_{n+1} = g(x_n)$ converges if the derivative condition holds at the fixed point:

$$|g'(p)| < 1 \implies \text{Convergent}, \quad |g'(p)| \geq 1 \implies \text{Divergent}$$

For $g(x) = x/2$: the fixed point is $p = 0$ and $g'(x) = 1/2$, so $|g'(0)| = 0.5 < 1$ — **convergent**.

► MATLAB Code

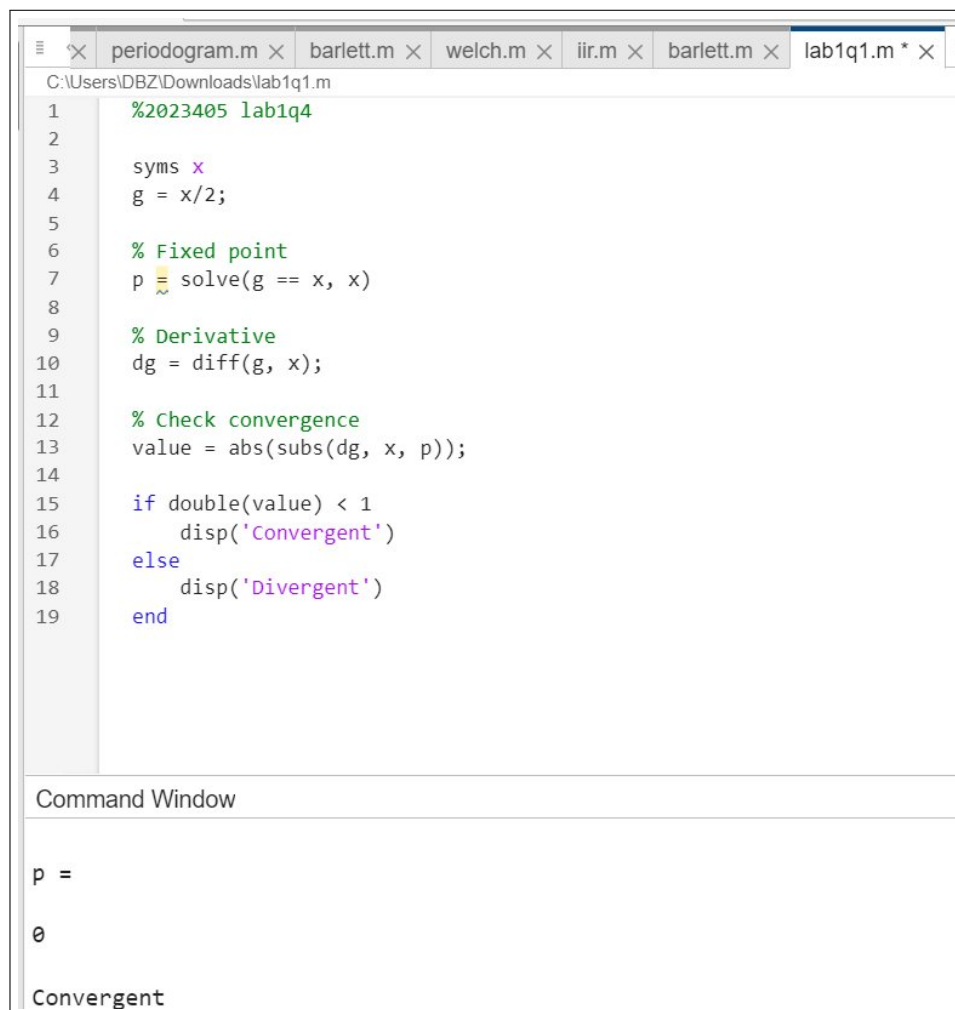
```

1  %2023405 lab1q4
2  syms x
3  g = x/2;
4
5  % Fixed point
6  p = solve(g == x, x)
7
8  % Derivative
9  dg = diff(g, x);
10
11 % Check convergence
12 value = abs(subs(dg, x, p));
13
14 if double(value) < 1

```

```
15     disp('Convergent')
16 else
17     disp('Divergent')
18 end
```

► Output



```
%2023405 lab1q4
syms x
g = x/2;

% Fixed point
p = solve(g == x, x)

% Derivative
dg = diff(g, x);

% Check convergence
value = abs(subs(dg, x, p));

if double(value) < 1
    disp('Convergent')
else
    disp('Divergent')
end
```

Command Window

```
p =
0
Convergent
```

Figure 4: MATLAB output for Task 04 — Convergence check for $g(x) = x/2$

► Discussion

MATLAB correctly identifies the fixed point as $p = 0$ and classifies the iteration as **Convergent**. Since $|g'(0)| = 0.5 < 1$, iterating from any starting point near 0 will cause the sequence to approach 0, confirming convergence by the fixed-point theorem.

5 Task 05 — Fixed-Point Divergence Check

► Question

For the iterative function $g(x) = 2x$, check whether the fixed-point iteration is **convergent** or **divergent**. Also iterate numerically starting from $x_0 = 1$ for 5 steps.

► Theory

For $g(x) = 2x$: the fixed point is $p = 0$ and $g'(x) = 2$, so $|g'(0)| = 2 > 1$. The iteration diverges — starting values are doubled at each step, causing the sequence to grow unboundedly: $1, 2, 4, 8, 16, 32, \dots$

► MATLAB Code

```

1  %2023405 q5
2  syms x
3
4  % Define the function
5  g = 2*x;
6
7  % Step 1: Find fixed point
8  p = solve(g == x, x);
9
10 % Step 2: Compute derivative
11 dg = diff(g, x);
12
13 % Step 3: Evaluate derivative at fixed point
14 value = abs(subs(dg, x, p));
15
16 disp('Fixed Point:'); disp(p);
17 disp('Derivative g''(x):'); disp(dg);
18 disp('|g''(p)| = '); disp(value);
19
20 % Step 4: Check convergence
21 if double(value) < 1
22     disp('The iteration is Convergent.');
```

```
26
27 % Step 5: Numerical iteration
28 g_num = @(x) 2*x;
29 x0 = 1;
30 N = 5;
31
32 vals = zeros(1, N+1);
33 vals(1) = x0;
34
35 for k = 1:N
36     vals(k+1) = g_num(vals(k));
37 end
38
39 disp('Iteration Values:');
40 disp(vals);
```

► Output

```

C:\Users\DBZ\Downloads\lab1q1.m
1 %2023405 q5
2 syms x
3
4 % Define the function
5 g = 2*x;
6
7 % Step 1: Find fixed point
8 p = solve(g == x, x);
9
10 % Step 2: Compute derivative
11 dg = diff(g, x);
12
13 % Step 3: Evaluate derivative at fixed point
14 value = abs(subs(dg, x, p));
15
16 disp('Fixed Point:');
17 disp(p);
18
19 disp('Derivative g'(x):');
20 disp(dg);
21
22 disp('|g'(p)| = ');
23 disp(value);
24
25 % Step 4: Check convergence
26 if double(value) < 1
27     disp('The iteration is Convergent.');
```

```

28 else
29     disp('The iteration is Divergent.');
```

```

30 end
31
32 % Step 5: Numerical iteration
33 g_num = @(x) 2*x;
34 x0 = 1;
35 N = 5;
36
37 vals = zeros(1, N+1);
38 vals(1) = x0;
39
40 for k = 1:N
41     vals(k+1) = g_num(vals(k));
42 end
43
44 disp('Iteration Values:');
45 disp(vals);

```

Command Window

```

Derivative g'(x):
2

|g'(p)| =
2

The iteration is Divergent.
Iteration Values:
    1     2     4     8    16    32

```

Figure 5: MATLAB output for Task 05 — Divergence check for $g(x) = 2x$

► Discussion

MATLAB correctly classifies $g(x) = 2x$ as **Divergent**. The derivative $|g'(p)| = 2 > 1$ violates the convergence condition. The numerical iteration starting from $x_0 = 1$ yields 1, 2, 4, 8, 16, 32 — values doubling each step, clearly diverging toward infinity. Contrasting this with Task 4 reinforces the significance of the derivative condition.

Conclusion

This lab demonstrated the power of MATLAB's Symbolic Math Toolbox for solving core calculus problems efficiently and accurately.

- **Task 1** verified that $\lim_{x \rightarrow \infty} (1 + 1/x)^x = e \approx 2.7183$ using symbolic limits.
- **Task 2** confirmed that the derivative of $e^{2x} + \sin(x)$ is $2e^{2x} + \cos(x)$ via the `diff()` function.
- **Task 3** computed the definite integral $\int_0^5 (x^2 + 7x) dx = 129.1667$, matching the analytical result.
- **Task 4** showed that $g(x) = x/2$ converges at its fixed point $p = 0$ since $|g'(0)| = 0.5 < 1$.
- **Task 5** demonstrated divergence for $g(x) = 2x$ since $|g'(0)| = 2 > 1$, producing an unbounded sequence.

Overall, MATLAB proved to be an effective tool for validating mathematical concepts, with results aligning precisely with hand-computed solutions.